

PRACTICAL-3

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).

- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

CODE:

```
import numpy as np
rows, cols = map(int, input().split())
elements = []
for _ in range(rows) :
    elements.extend(map(int, input().split()))
array = np.array(elements).reshape(rows, cols)
print(array)
print(array.ndim)
print(array.shape)
print(array.size)
```

The screenshot shows the CODETANTRA online code editor interface. At the top, there's a navigation bar with 'Home' and 'Learn Anywhere' links. The user's email '202501050007@mitace.ac.in' and a 'Logout' button are visible on the right. Below the navigation bar, a status bar indicates '2 out of 2 shown test case(s) passed' and '10 out of 10 hidden test case(s) passed'. The main area displays two test cases. Test case 1 shows 'Expected output' and 'Actual output' sections with identical results: a 4x4 array of integers. Test case 2 shows 'Expected output' and 'Actual output' sections with identical results: a 1x1 array containing the integer 0. At the bottom, there's a 'Terminal' and 'Test cases' tab, and a navigation bar with '< Prev', 'Reset', 'Submit', and 'Next >' buttons.

3.2.1. Numpy: Matrix Operations

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition ()**
2. **Subtraction ()**
3. **Element-wise Multiplication ()**
4. **Matrix Multiplication ()**
5. **Transpose of Matrix A Input Format:**
 - The user will input 3 rows for , each containing 3 integers separated by spaces.
 - Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

CODE

```

import numpy as np # Input matrices print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Matrix B:") matrix_b = np.array([list(map(int,
input().split())) for i in range(3)])

# Addition addition_result = matrix_a + matrix_b
print("Addition (A + B):") print(addition_result) #
Subtraction subtraction_result = matrix_a - matrix_b
print("Subtraction (A - B):") print(subtraction_result) #
Multiplication (element-wise)
elementwise_multiplication_result = matrix_a * matrix_b

print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result) # Matrix
multiplication (dot product)
matrix_multiplication_result=np.dot(matrix_a, matrix_b)
print("A dot B:")

print(matrix_multiplication_result)

# Transpose transpose_a
= matrix_a.T
print("Transpose of A:")

```

```
print(transpose_a)
```

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).

- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

CODE:

```
import numpy as np # Input matrices print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])
print("Enter Array2:") arr2 = np.array([list(map(int,
input().split())) for i in range(3)])
# Perform horizontal stacking (hstack)
a=np.hstack((arr1,arr2))
print("Horizontal Stack:") print(a)
# Perform vertical stacking (vstack)
b=np.vstack((arr1,arr2)) print("Vertical
```

Stack:")

The image displays two screenshots of the CODETANTRA online IDE interface, showing the results of a program execution for a stack problem. The interface includes a top navigation bar with the CODETANTRA logo, a home button, and a user profile section. The main area is divided into a left sidebar with an 'Explorer' tab and a central workspace. The workspace shows the 'Test case 1' and 'Test case 2' results, including the expected output, actual output, and the state of the horizontal and vertical stacks. The test cases are marked as 'passed'.

Test case 1:

- Expected output: `1 2 3`, `4 5 6`, `7 8 9`
- Actual output: `1 2 3`, `4 5 6`, `7 8 9`
- Horizontal stack: `[[1, 2, 3, 4, 5, 6]]`, `[-4, -5, -6, -7, -8, -9]`, `[-7, -8, -9, 10, 11, 12]]`
- Vertical stack: `[[1, 2, 3]]`, `[-4, -5, -6]`, `[-7, -8, -9]`, `[-4, -5, -6]`, `[-7, -8, -9]`, `[-10, -11, -12]]`

Test case 2:

- Expected output: `-1 -2 -3`, `-4 -5 -6`
- Actual output: `-1 -2 -3`, `-4 -5 -6`
- Horizontal stack: `[[1, 2, 3, 4, 5, 6]]`, `[-4, -5, -6, -7, -8, -9]`, `[-7, -8, -9, 10, 11, 12]]`
- Vertical stack: `[[1, 2, 3]]`, `[-4, -5, -6]`, `[-7, -8, -9]`, `[-4, -5, -6]`, `[-7, -8, -9]`, `[-10, -11, -12]]`

3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.

- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

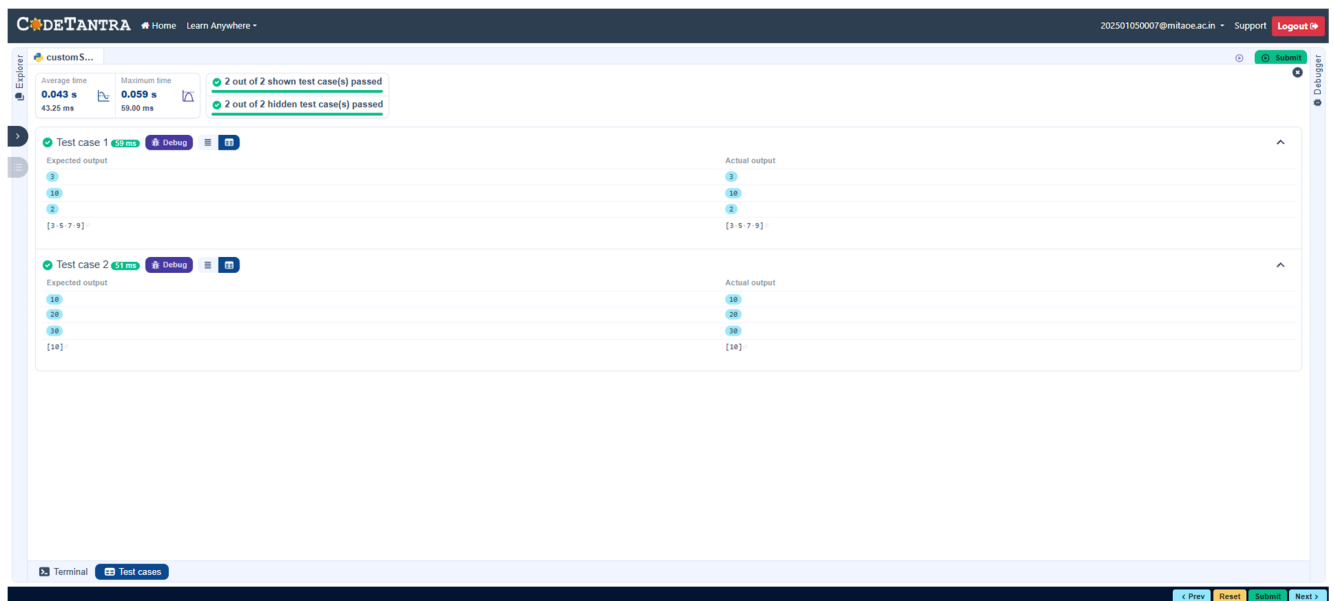
CODE:

```
import numpy as np

# Take user input for the start, stop, and step of the sequence
start = int(input()) stop = int(input()) step = int(input())

# Generate the sequence using np.arange()
a= np.arange(start,stop, step) print(a)

# Print the generated sequence
```

3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.

- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

CODE:

```
import numpy as np
def array_operations(A, B):
    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)    # Arithmetic
    Operations
    sum_result = A + B
    diff_result = A - B    prod_result = A
    * B
    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A) std_dev_A
    = np.std(A)
    # Bitwise Operations
    and_result = A & B
    or_result = A | B
    xor_result
```

```

# Output results with one space between each element
print("Element-wise Sum:", ' '.join(map(str, sum_result)))
print("Element-wise Difference:", ' '.join(map(str, diff_result)))
print("Element-wise Product:", ' '.join(map(str, prod_result)))
print(f"Mean of A: {mean_A}")

print(f"Median of A: {median_A}")

print(f"Standard Deviation of A: {std_dev_A}")

print("Bitwise AND:", ' '.join(map(str, and_result)))
print("Bitwise OR:", ' '.join(map(str, or_result)))
print("Bitwise XOR:", ' '.join(map(str, xor_result))) A =
list(map(int, input().split())) # Elements of array A B =
list(map(int, input().split())) # Elements of array B
array_operations(A, B)

```

The screenshot shows the CodeTANTRA IDE interface. At the top, there's a navigation bar with 'Home' and 'Learn Anywhere' links. The status bar indicates '202501050007@mitae.ac.in' and a 'Logout' button. The main area displays test results for 'Test case 1'. The 'Expected output' and 'Actual output' sections both show the same results, indicating a successful test case run. The results include element-wise sum, difference, product, mean, median, standard deviation, and bitwise operations for arrays A and B.

Expected output	Actual output
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

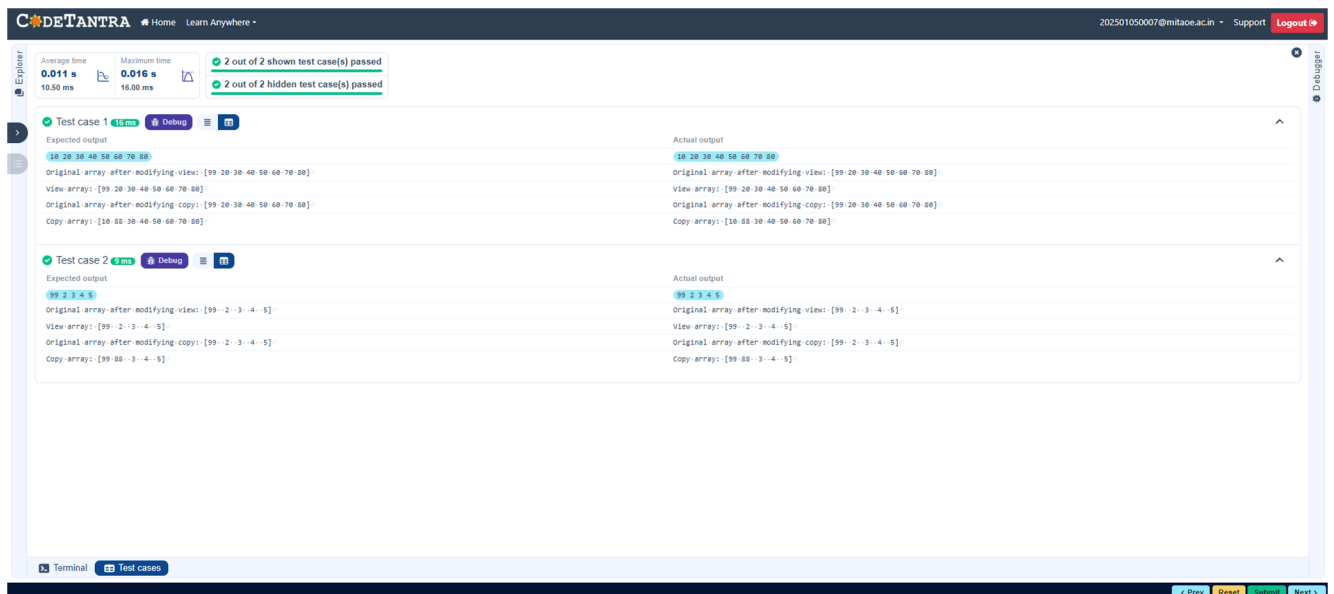
Copy array: <copy_array> **CODE:**

```
import numpy as np
inputlist = list(map(int,input().split(" "))
)# Original array original_array =
np.array(inputlist) # Create a view
view_array =original_array.view()
```

```
# Create a copy copy_array
=original_array.copy()

# Modify the view view_array[0] = 99 print("Original array
after modifying view:", original_array) print("View array:",
view_array)

# Modify the copy copy_array[1] = 88 print("Original array
after modifying copy:", original_array) print("Copy array:",
copy_array)
```



3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.

- `count_value`: The value to count its occurrences in the array.
- `broadcast_value`: The value to add for broadcasting across the array. You need to complete the code to perform the following operations:

1. Searching: Find the indices where `search_value` appears in `array1` and print these indices.

2. Counting: Count how many times `count_value` appears in `array1` and print the count.

3. Broadcasting: Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.

4. Sorting: Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

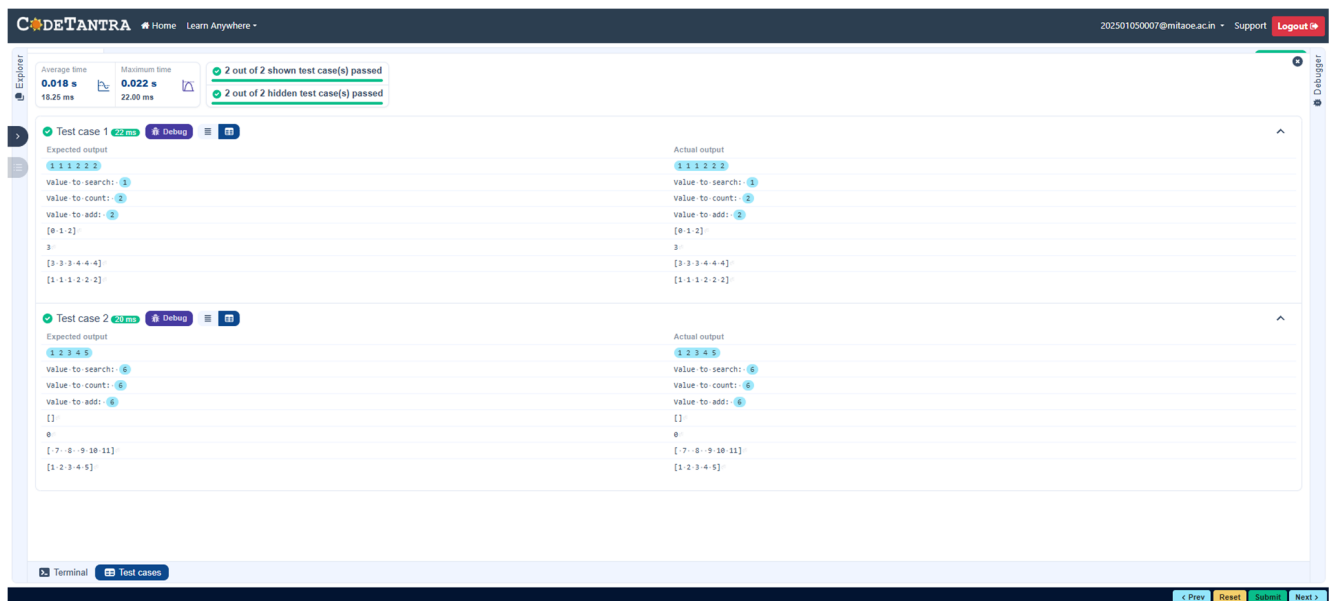
Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.

4. The sorted array.

CODE:

```
import numpy as np# Input array from the user array1 =
np.array(list(map(int, input().split()))))
# Searching search_value = int(input("Value to
search: ")) count_value = int(input("Value to
count: ")) broadcast_value = int(input("Value
to add: ")) # Find indices where value matches
in array1
a=np.where(array1==search_value)[0] print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value) print(b)
# Broadcasting addition c=array1 +
broadcast_value print(c)
# Sort the first array
d=np.sort(array1) print(d)
```



3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.

- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.

- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

CODE:

```
import numpy as npa = np.loadtxt("Sample.csv", delimiter=',',
skiprows=1)
```

```
# 1. Print all student details print("All
student Details:\n",a)
```

```
# 2. print total students
```

```
print("Total Students:",a.shape[0]) # 3. Print all
```

```
student Roll numbers print("All Student Roll
```

```
Nos",a[:,0]) # 4. Print subject 1 marks
```

```

print("Subject 1 Marks",a[:,1]) # 5. print
minimum marks of Subject 2 print("Min marks
in Subject 2",np.min(a[:,2])) # 6. print
maximum marks of Subject 3 print("Max marks
in Subject 3",np.max(a[:,3]))
# 7. Print All subject marks print("All subject
marks:",a[:,1:4]) # 8. print Total marks of
students print("Total
Marks",np.sum(a[:,1:4],axis=1))
# 9. print average marks of each student
#Note:Expected output shows rounded values like 77.3
print(np.round(np.mean(a[:,1:4],axis=1),1)) # 10. print
average marks of each subject print("Average Marks of
each
subject",np.round(np.mean(a[:,1:4],axis=0),1))
# 11. print average marks of S1 and S2
#The test case expected[71.7 59.8], which are the individual subject
averages
print("Average Marks of S1 and
S2",np.round(np.mean(a[:,1:3],axis=0),1)) # 12. print average
marks of S1 and S3 print("Average Marks of S1 and
S3",np.round(np.mean(a[:,[1, 3]],axis=0),1))

```

```

# 13. print Roll number who got maximum marks in Subject 3
print("Roll no who got maximum marks in Subject
3",a[np.argmax(a[:,3]),0])

# 14. print Roll number who got minimum marks in Subject 2
print("Roll no who got minimum marks in Subject
2",a[np.argmin(a[:,2]),0])

# 15. print Roll number who got 24 marks in Subject 2 print("Roll no
who got 24 marks in Subject 2",a[a[:,2]==24][:,0:1]) # 16. print
count of students who got marks in Subject 1 < 40 print("Count of
students who got marks in Subject 1 < 40",np.sum(a[:,1]<40))

# 17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 >
90:",np.sum(a[:,2]>90))

# 18. print count of students in each subject who got marks >= 90
print("Count of students in each subject who got marks >=
90:",np.sum(a[:,1:4]>=90,axis=0))# 19. print count of subjects in
which each student got marks >= 90 print("Roll no:",a[:,0])
print("Count of subjects in which student got marks >=
90:",np.sum(a[:,1:4]>=90,axis=1)) # 20. Print S1 marks in
ascending order print(np.sort(a[:,1]))

# 21. Print S1 marks >= 50 and <=
90print(a[(a[:,1]>=50)&(a[:,1]<=90)])

#Bonus: The test case shows the full array printed again before
the index positions print(a)

```

22. Print the index position of marks 79

Expected format:(array([6,9],dtype=int32),)

print(np.where(a[:,1]==79))

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Explorer Average time 0.029 s Maximum time 29.00 ms 1 out of 1 shown test case(s) passed

Test case 1 69 ms Debug

Expected output

All student details:-

```
[[381.-, 67.-, 77.-, 88.-]
 [382.-, 78.-, 88.-, 77.-]
 [383.-, 45.-, 56.-, 89.-]
 [384.-, 88.-, 98.-, 45.-]
 [385.-, 78.-, 88.-, 99.-]
 [386.-, 68.-, 78.-, 36.-]
 [387.-, 79.-, 12.-, 45.-]
 [388.-, 46.-, 45.-, 77.-]
 [389.-, 89.-, 32.-, 88.-]
 [318.-, 79.-, 24.-, 33.-]]
```

Total Students: 10

All Student Roll Nos [381, 382, 383, 384, 385, 386, 387, 388, 389, 318.]

Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]

Min marks in Subject 2 12.0

Max marks in Subject 3 99.0

All subject marks: [[67, 77, 88.]

```
[78, 88, 77.]
 [45, 56, 89.]
 [88, 98, 45.]
 [78, 88, 99.]
 [68, 78, 36.]
 [79, 12, 45.]
 [46, 45, 77.]
 [89, 32, 88.]
 [79, 24, 33.]]
```

Total Marks [232, 242, 190, 231, 265, 182, 136, 168, 209, 136.]

```
[77.8 81.-, 63.3 77.-, 88.8 68.7 45.3 56.-, 69.7 45.3]
```

Actual output

All student details:-

```
[[381.-, 67.-, 77.-, 88.-]
 [382.-, 78.-, 88.-, 77.-]
 [383.-, 45.-, 56.-, 89.-]
 [384.-, 88.-, 98.-, 45.-]
 [385.-, 78.-, 88.-, 99.-]
 [386.-, 68.-, 78.-, 36.-]
 [387.-, 79.-, 12.-, 45.-]
 [388.-, 46.-, 45.-, 77.-]
 [389.-, 89.-, 32.-, 88.-]
 [318.-, 79.-, 24.-, 33.-]]
```

Total Students: 10

All Student Roll Nos [381, 382, 383, 384, 385, 386, 387, 388, 389, 318.]

Subject 1 Marks [67, 78, 45, 88, 78, 68, 79, 46, 89, 79.]

Min marks in Subject 2 12.0

Max marks in Subject 3 99.0

All subject marks: [[67, 77, 88.]

```
[78, 88, 77.]
 [45, 56, 89.]
 [88, 98, 45.]
 [78, 88, 99.]
 [68, 78, 36.]
 [79, 12, 45.]
 [46, 45, 77.]
 [89, 32, 88.]
 [79, 24, 33.]]
```

Total Marks [232, 242, 190, 231, 265, 182, 136, 168, 209, 136.]

```
[77.8 81.-, 63.3 77.-, 88.8 68.7 45.3 56.-, 69.7 45.3]
```

< Prev Reset Submit Next >

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Explorer Average time 0.029 s Maximum time 29.00 ms 1 out of 1 shown test case(s) passed

Average Marks of each subject [71.7 59.8 67.7]

Average Marks of S1 and S2 [71.7 59.8]

Average Marks of S1 and S3 [71.7 67.7]

Roll no who got maximum marks in Subject 3 385.0

Roll no who got minimum marks in Subject 2 387.0

Roll no who got 24 marks in Subject 1 [318.]

Count of students who got marks in Subject 1 < 40 0

Count of students who got marks in Subject 2 > 90: 1

Count of students in each subject who got marks >= 90: [0 1 1]

Roll no: [381, 382, 383, 384, 385, 386, 387, 388, 389, 318.]

Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]

```
[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]
 [[381.-, 67.-, 77.-, 88.-]
 [382.-, 78.-, 88.-, 77.-]
 [383.-, 45.-, 56.-, 89.-]
 [384.-, 88.-, 98.-, 45.-]
 [385.-, 78.-, 88.-, 99.-]
 [386.-, 68.-, 78.-, 36.-]
 [387.-, 79.-, 12.-, 45.-]
 [388.-, 46.-, 45.-, 77.-]
 [389.-, 89.-, 32.-, 88.-]
 [318.-, 79.-, 24.-, 33.-]]
```

Average Marks of each subject [71.7 59.8 67.7]

Average Marks of S1 and S2 [71.7 59.8]

Average Marks of S1 and S3 [71.7 67.7]

Roll no who got maximum marks in Subject 3 385.0

Roll no who got minimum marks in Subject 2 387.0

Roll no who got 24 marks in Subject 1 [318.]

Count of students who got marks in Subject 1 < 40 0

Count of students who got marks in Subject 2 > 90: 1

Count of students in each subject who got marks >= 90: [0 1 1]

Roll no: [381, 382, 383, 384, 385, 386, 387, 388, 389, 318.]

Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]

```
[45, 46, 67, 68, 78, 78, 79, 79, 88, 89.]
 [[381.-, 67.-, 77.-, 88.-]
 [382.-, 78.-, 88.-, 77.-]
 [383.-, 45.-, 56.-, 89.-]
 [384.-, 88.-, 98.-, 45.-]
 [385.-, 78.-, 88.-, 99.-]
 [386.-, 68.-, 78.-, 36.-]
 [387.-, 79.-, 12.-, 45.-]
 [388.-, 46.-, 45.-, 77.-]
 [389.-, 89.-, 32.-, 88.-]
 [318.-, 79.-, 24.-, 33.-]]
```

< Prev Reset Submit Next >